

Week 2 Recitation

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ECON2001.01 (Prin. Micro.)

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TA Office Hours

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 - Feel free to reach out!
 - Please indicate in the subject field:
"*[ECON2001.01, F 8:00AM]* ~ "
This would help me get to you quickly.
- **Office hour:** Mon **1:30 - 2:30PM**, Wed **2:30 - 3:30PM** | 📍Arps 310

Review

<Optimization: Doing the best you can>

Optimization using Total values

- Optimum, Net Benefit, Cost-Benefit analysis

Optimization using Marginal analysis

- Marginal Benefit, Marginal Cost, Principle of Optimization at the Margin

Review

How to optimize?

There are two methods in evaluating what is the optimal choice:

- Optimization using **total values**

The optimum is determined by calculating net-benefits for all alternatives and choosing the alternative that has the highest net-benefit with the optimization

- Optimization using **marginal analysis**

The optimum is determined by comparing if one is better off moving towards an alternative and worse off when moving away with the optimization

Importantly, using either method will always get you to the **same optimal choice**.

(The optimum is always the one that maximizes Net-Benefits.)

Review

Example

Quantity	Total Benefit	Marginal Benefit (MB)	Total Cost	Marginal Cost (MC)	Net Benefit
0	0	-	0	-	0
1	10		3		
2	18		11		

- Possible choices (alternatives)?
- Total value analysis?
- Marginal analysis?
 - $MB = MC$?

Review

Example

What if $MB \neq MC$ across all alternatives?

Review

Example

What if $MB \neq MC$ across all alternatives?

Quantity	Total Benefit	Marginal Benefit (MB)	Total Cost	Marginal Cost (MC)	Net Benefit
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1	10		3		
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- Possible choices (alternatives)?
- Total value analysis?
- Marginal analysis?

Review

Principle of Optimization at the Margin

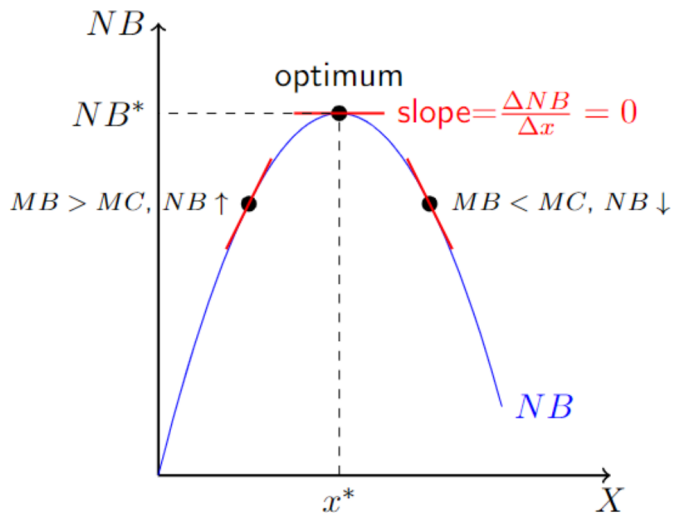
Principle of Optimization at the Margin:

An optimal feasible alternative has the property that

- moving to it makes you better off and
- moving away from it makes you worse off

Review

Graphic illustration



Review

Example - Allocating time

You have 3 possible hours to study for Chemistry or Biology. How will you allocate your time between the two subjects which will **maximize the score** you will get on the next day's exams?

Review

Example - Allocating time

You have 3 possible hours to study for Chemistry or Biology. How will you allocate your time between the two subjects which will **maximize the score** you will get on the next day's exams?

Hour(s)	Chemistry (Score)	Marginal Score in Chem	Biology (Score)	Marginal Score in Bio
0	65	-	64	-
1	72		72	
2	77		78	
3	80		82	

- Benefits? Costs?
- Your goal?
- Total value analysis?
- Marginal analysis?

Problem Set - Exercise

Chapter 3

1. (Optimization in Total Values) Suppose there are three possible airports near your home that you use for a flight to Boston. At the closest airport you find an airfare of \$400, the middle airport (the second closest to your home) has an airfare of \$325, and you would pay \$275 to fly from the farthest airport. Assume you want to take a taxi to the airport and the roundtrip cost to the close airport is \$20 and will take 15 minutes. The taxi fare to the second airport is \$30 and will take 30 minutes. The taxi fare to the farthest airport is \$60 and will take one hour. You have a part-time job where you earn \$12 per hour. Assume the benefit flying is the same for each airport. Create a table similar to Apartment example presented during lecture and determine which airport is the optimum for you to use.

Problem Set - Exercise

Chapter 3

2. (Optimization in Total Values and Marginal Analysis) Suppose you and your group are wanting to start a business. One of the decisions that you must make, is to determine how much to produce. The below table provides the benefits and costs associated with each of the quantities shown. Refer to the below table. Deriving marginal benefit, marginal cost, and the net-benefit for each of the quantities shown, determine the optimum. Explain how you arrive your answer using both optimization techniques—total values and marginal analysis.

Table 1: **Production Information**

Quantity	Total Benefit	Marginal Benefit	Total Cost	Marginal Cost	Net-Benefit
0	0	—	0	—	
10	300		25		
20	600		100		
30	900		225		
40	1200		400		
50	1500		625		
60	1800		900		
70	2100		1225		
80	2400		1600		

Problem Set - Exercise

Chapter 3

3. (Optimization using Marginal Analysis) Suppose that you're trying to decide how many pieces of sushi to buy and eat. Sushi costs \$2 per piece and the benefit of consuming each piece of sushi, i.e., marginal benefit, is

$$MB = \$24 - \$2x$$

where x denotes the number of sushi pieces. Determine the optimal amount of sushi to purchase. Explain your answer.